

See discussions, stats, and author profiles for this publication at: <https://www.researchgate.net/publication/384676546>

Students' behavior mining in e-learning environment using cognitive processes with information technologies

Article in Education and Information Technologies · September 2019

DOI: 10.1007/s10639-019-09892-5

CITATIONS

296

READS

212

2 authors:



Ahmad Jalal

Pohang University of Science and Technology

631 PUBLICATIONS 22,944 CITATIONS

SEE PROFILE



Maria Mahmood

Rawalpindi Women University

11 PUBLICATIONS 1,494 CITATIONS

SEE PROFILE



Students' behavior mining in e-learning environment using cognitive processes with information technologies

Ahmad Jalal¹ · Maria Mahmood¹

Received: 7 December 2018 / Accepted: 25 February 2019 / Published online: 13 March 2019
© Springer Science+Business Media, LLC, part of Springer Nature 2019

Abstract

Rapid growth and recent developments in education sector and information technologies have promoted E-learning and collaborative sessions among the learning communities and business incubator centers. Traditional practices are being replaced with webinars (live online classes) E-Quizes (online testing) and video lectures for effective learning and performance evaluation. These E-learning methods use sensors and multimedia tools to contribute in resource sharing, social networking, interactivity and corporate trainings. While, artificial intelligence tools are also being integrated into various industries and organizations for students' engagement and adaptability towards the digital world. Predicting students' behaviors and providing intelligent feedbacks is an important parameter in the E-learning domain. To optimize students' behaviors in virtual environments, we have proposed an idea of embedding cognitive processes into information technologies. This paper presents hybrid spatio-temporal features for student behavior recognition (SBR) system that recognizes student-student behaviors from sequences of digital images. The proposed SBR system segments student silhouettes using neighboring data points observation and extracts co-occurring robust spatio-temporal features having full body and key body points techniques. Then, artificial neural network is used to measure student interactions taken from UT-Interaction and classroom behaviors datasets. Finally a survey is performed to evaluate the effectiveness of video based interactive learning using proposed SBR system.

Keywords Artificial intelligence · Behavior mining · E-learning · Student behavior recognition

✉ Ahmad Jalal
ahmadjalal@mail.au.edu.pk

Maria Mahmood
maria.mehmood@mail.au.edu.pk

¹ Department of Computer Science and Engineering, Air University, E-9, Islamabad, Pakistan

1 Introduction

Learning is a domain ruled by teacher-student interactions or learner-learner interactions. Artificial intelligence is changing the E-learning industry as smart simulations, intelligent tutoring systems, virtual mentors and interactive learning management systems. Student behavior recognition (SBR) has gained much importance in recent years and has become a challenging topic to investigate by the research community (Buys et al. 2014; Oberg et al. 2012). Usually, behavior interactions are body-language, facial expressions, and feelings between multiple persons. However, distinguishing such behavior interactions in real environment is still a challenging task because of lighting conditions, clothing colors, partial occlusions, and other pedestrians involved in a scene (Jalal and Zeb 2008; Kanungo et al. 2002; Yang and Tian 2014). Therefore, dealing with various research issues, this research topic has received valuable utilization and considerable attention by many real-world applications such as E-learning (Sehar et al. 2018), security (Yang and Tian 2012), human motion tracking (Jalal et al. 2014; Muller and Roder 2006), pattern recognitions (Mahmood et al. 2018), behavior mining (Fatahi et al. 2018; Aissaoui et al. 2018), smart surveillance (Zhao et al. 2013) and intelligent video retrievals (Jalal et al. 2013).

In the behavior recognition field, some limited researchers have contributed their ideas based on different features techniques. For instance, Houda and Yannick (2014) described visual words using 3D SIFT feature extractor and classified interactions based on co-occurrence of visual words using K-nearest neighbor as recognizer engine. They achieved reasonable performance, however, selection of number of visual words affected their recognition rate. In Ryoo and Aggarwal (2009), designed a spatio-temporal relationship match to measure structural similarity among video sequences. They used a set of predicates to describe spatial and temporal relationships among feature points and obtained excellent success rate. In Berlin and John (2016), used Harris corner detector as feature extractor and histogram of region of interest to represent each interaction. They assumed static background and thus failed to cope with real environment. In Chattopadhyay and Das (2016), defined interest points by considering significant variations in intensity values and motion. They represented each interest point using a set of feature descriptors, however, still observed some inaccuracy due to the inappropriate detection of interest points and intra-class variations in appearance of different actors. In Zhan and Chang (2014), adopted a pictorial structures model to locate the region of human body parts and describe relations between them based on relative distance, relative velocity, individual velocity, hand distance, foot height, and area of lower body. However, it is still difficult to depend upon above methodologies, because they mainly rely on local features, static backgrounds, simple interactions or pose estimations. Therefore, we needed to develop a novel framework which provides robust spatio-temporal features for SBR.

In this paper, we proposed a new hybrid spatio-temporal features approach for behavior mining based on full body silhouettes and individual body points of E-learners. The methodology undergoes following series of steps. As the first step, student silhouettes are detected in video sequences based on height and width of spatially connected components and segmented from background using a threshold parameter. In the second step, the key body points of individual silhouette are retrieved

using centroids of actors as scaling parameter. During hybrid feature extraction techniques, intensity changes are observed across all the frames to describe temporal relationships, while, distance between key body points is estimated to define spatial relationships among them. At final step, co-occurring spatio-temporal features determine student-student interaction through artificial neural network (ANN) over UT-Interaction and classroom behaviors dataset. UT-Interaction dataset is divided into two sets. Set 1 is captured in a parking lot, while set 2 is taken in a windy lawn. Each set comprises of 10 video sequences, with each video containing 6 human interactions. However, classroom behaviors dataset contains 12 student interactions in classroom environment. Our approach performs well on both the datasets giving significant recognition accuracy over other state-of-the-art methods, respectively.

The major contributions of this paper are summarized as follow: (1) We have proposed an idea of recognizing student behaviors in the E-learning environment (2) Classroom behaviors dataset is collected after introducing video based interactive learning among kindergarten students. (3) We introduced hybrid spatio-temporal features approach having combination of full body silhouettes and key body points for feature extraction. (4) Also, our pre-processing model can distinguish critical behavior interactions between students. (5) The proposed SBR system is normalized first to provide invariant characteristics. (6) Based on our certain knowledge, it is the first time to cope with SBR field and E-learning using UT-Interaction and classroom behaviors dataset. (7) A web application is designed for martial arts training in virtual environments (8) A questionnaire based survey is performed to evaluate the effectiveness of proposed idea.

The rest of the paper is arranged as follows: Section 2 presents related works on behavior recognition, Section 3 presents an overview of the solution framework which comprises of silhouette capturing and representation, key body points retrieval, feature extraction, codebook design and student behavior recognition. Section 4 presents the results of proposed SBR methodology using UT-Interaction and classroom behaviors datasets. Finally, Section 5 concludes the paper.

2 Related work

During the past decade, student behavior recognition has diverted the attention of a lot of researchers in computer vision and E-learning community. For instance, Sherlock is an intelligent tutoring system being used to teach airforce technicians to diagnose electrical system problems in aircraft. Avatar-based training modules developed by the University of Southern California to train military personnel being sent on international posts is another example of intelligent tutoring systems. In Hwang et al. (2009), a context aware u-learning environment is developed for guiding inexperienced researchers to practice single crystal X-ray diffraction operations. In Lu et al. (2012) human activity recognition is applied in educational domain, resulting in ubiquitous learning, in which the system can detect the students' behavior and provide personalized support to guide the students to learn in real world. Speech emotion recognition (Chen et al. 2007), gesture recognition (Kowalewski et al. 2013), and action recognition (Zaletelj and Košir 2017) have already been explored in the

E-learning domain for predicting various learning styles of the students. In Sabanc and Bulut (2018), behavior recognition and behavior management of the students in the inclusive education environment with talented and gifted pupils for social emotional development is presented. There are some students who push and shove others, can never wait their turn, mistreat the materials in the classroom and generally create chaos wherever they go. These students aren't challenging, but they may have challenging behaviors. Recognizing their behaviors as positive and negative can help determining their learning needs, configuring education in accordance with learning styles, revealing their interests, and abilities by enhancing students participation in education process. In the next subsections, we have categorized the behavior interactions into three classes as; student-robot interaction, student-object interaction and student-student interaction.

2.1 Student-robot interaction

Robots that interact with learners need to understand the social behaviors of learners to perform socially appropriate interactions in response. For an instance, Gaschler et al. (2012) designed the bartender robot of the project JAMES with the aim of interpreting non-verbal cues of human customers and hand out ordered beverages. They used head pose and body posture recognition to analyze the most important gestures of the customers in a bar scenario. Then, they captured spatial arrangement in a group to express engagement in interaction. Similarly, Fujii et al. (2014) developed a service robot operated by human gestures for an office environment. In their system, user can give orders to the robot through predefined gestures. After recognition of the user's commands in real time, the robot replies with an audio/visual message and starts the service tasks demanded by the users. The idea of gesture recognition is elaborated to explore physical student activities and behavior recognition. Recently, activity recognition is integrated in several consumer products such as game consoles, personal fitness training applications, health monitoring, fall detection, smart homes and self-managing systems. Several probability-based algorithms have been used to build activity models. In Babiker et al. (2017), developed an automated daily human activity recognition system for video surveillance. They used robust neural network and multi-layer feed forward perceptron network to classify activities model. Different techniques have been proposed for simple activities recognition. However, recognition of complex activities involving interactions with other persons or objects is still a challenging and most demanding task nowadays.

2.2 Student-object interaction

To understand student-object interaction in the visual world, automatic recognition of individual students and objects is not enough. Therefore, structured relationships have to be associated between them to analyze the ongoing behavior interactions. For instance, Gkioxari et al. (2018) aimed to detect (human, verb, object) triplets in challenging everyday photos. They used appearance of a person as a cue to recognize the action and localize the interacted objects. Then, the action specific density over targeted object location is estimated and fused with human action to infer the

human-object interaction. In Shen et al. (2018) proposed a factorized model for human-object interaction recognition that decomposes reasoning on previously seen verbs and objects to produce novel verb-object pairs. They trained a full joint model to locate and score all verb as well as object classes. During testing, scores for all combinations of verb-object pairs are computed to predict the final human-object interaction.

2.3 Student-student interaction

Nowadays, monitoring and recognition of complex social interactions occurring between learners become a real issue in the field of computer vision because of partial and full occlusions of the target persons at public and crowded areas. Researchers are mainly focused on appearance and geometric information using RGB and depth video sequences to expand the flexibility of recognition under real environments. For instance, Cho et al. (2017) designed compositional interaction descriptors for

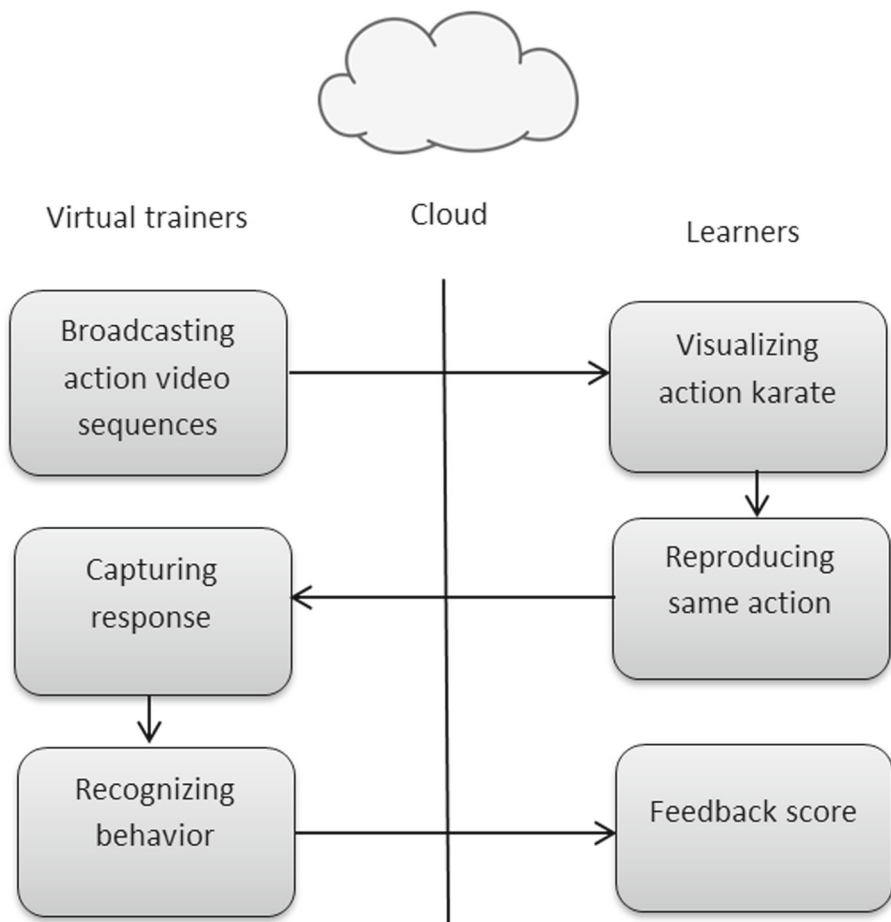


Fig. 1 Proposed system architecture for training martial arts in E-learning environment

human interaction recognition via RGB image sequences. These descriptors represent motion relationships on different levels to provide a high discriminative power for complex interactions. They combined individual descriptors into compositional descriptor for interaction representation. To avoid motion ambiguities in interactions, Kong et al. (2014) observed interdependencies at the action level and body part level. A multi-class Adaboost is proposed to select discriminative body parts of interacting persons. Then, their interdependencies are discovered using minimum spanning tree algorithm. Lastly, local features of body parts are fused with global features of individual actions to cope with complicated human interactions.

3 Overview of solution framework

The proposed SBR system is designed to evaluate behaviors of junior karate learners in E-learning environment for improving their flexibility, balance and strength. Karate is an Asian system of unarmed combat using the hands and feet to deliver and block blows, widely practiced as a sport. The system helps kindergarten students in increasing their self-esteem and body image by giving feedbacks on their actions. The idea is to provide online Martial Arts training via virtual trainers or action videos followed by students response reproducing the same action. The evaluation of learners is carried out by recognizing their behaviors and providing feedback scores for their motivation. An overview of system methodology is depicted in Fig. 1. The system operates in an online setting. Training starts by playing martial arts videos for the learners. Learners visualize the depicted actions and perform the same behavior interactions. The system then captures their actions and recognize their behaviors.

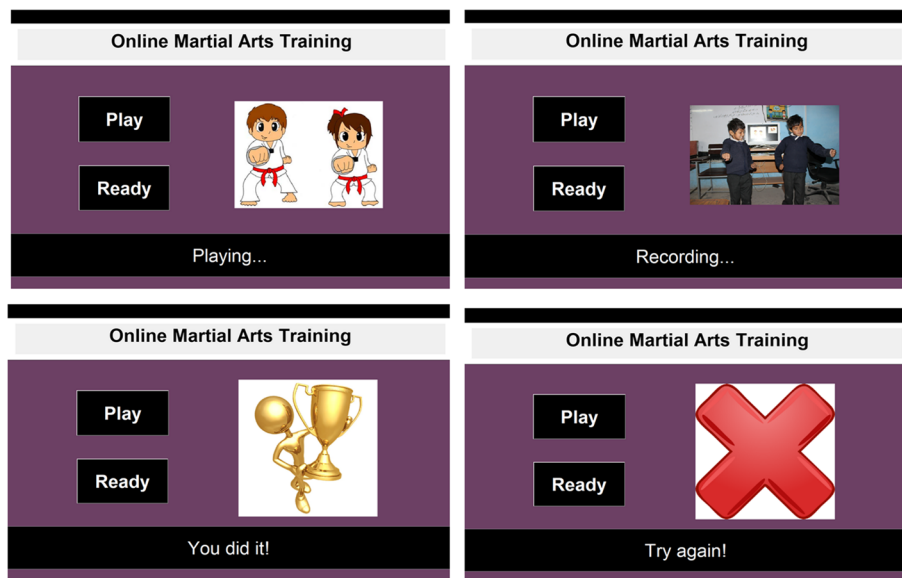


Fig. 2 Online video based interactive environment for martial arts training

Based on predicted behaviors and actual videos identifiers, feedback scores are produced. Figure 2 shows a platform designed for pre-school kids to learn martial arts. The application's graphical user interface contains two buttons and a text display. Pressing 'play' randomly plays any of the learned interactions. While 'ready' button starts capturing the scene. If the captured interaction matches to that displayed, winner is declared. However, if the user fails to perform the same, try again message is displayed. Detailed description of behavior recognition process is shown in Fig. 3. The process is divided into two modules as training and testing while each module is subdivided into four phases. Initially, silhouette capturing and representation phase involves detection of student silhouettes in video sequences and their segmentation from the background. Secondly, key body points retrieval phase involves marking of essential points at boundary of individual silhouettes. Thirdly, hybrid feature extraction phase involves description of full silhouette based temporal relationships and

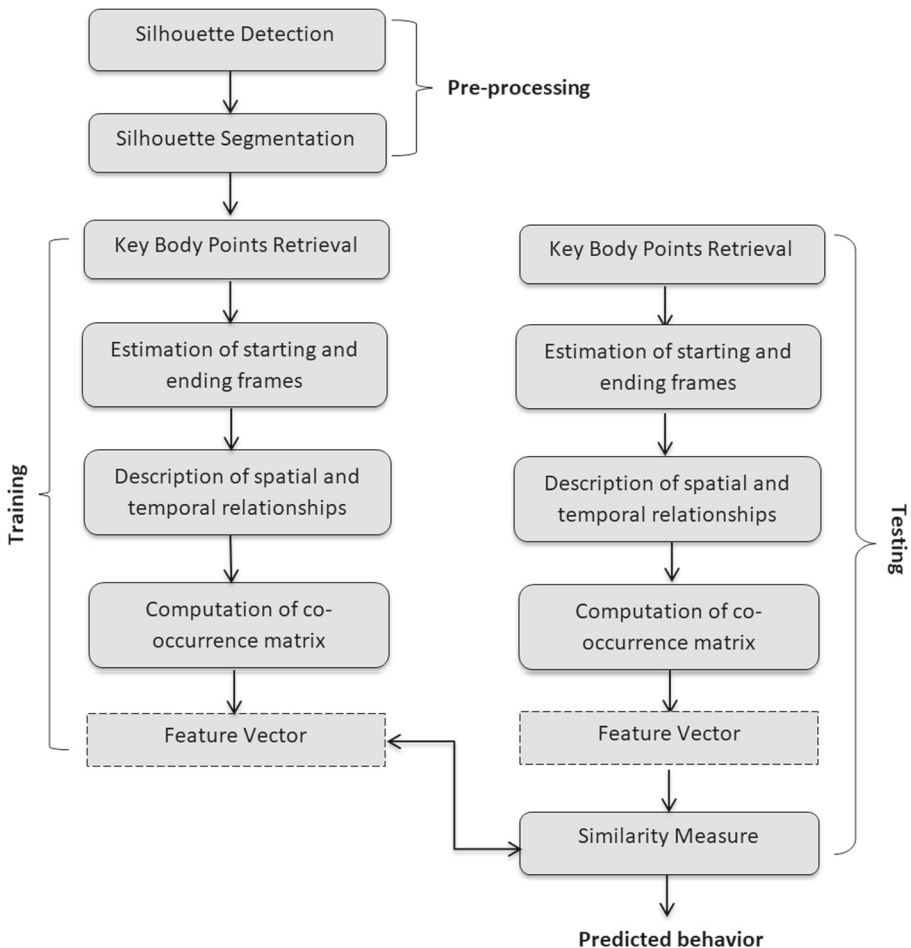


Fig. 3 Process flow for Student Behavior Recognition of martial arts learners

key body points based spatial relationships. Lastly, student behaviors are recognized by classification of input video sequences to their belonging classes.

3.1 Silhouette capturing and representation

The key objective of silhouettes capturing and representation is to extract an effective silhouette generation of the student-student behavior interactions from noisy backgrounds. Therefore, we fixed image size upto 300 x 200 and applied denoised effects using averaging filter algorithm (Ma et al. 2009; Jalal et al. 2012; Milanfar 2012). It provides smooth data by replacing each data point with the average of the neighboring data points defined within the span. Auto-bounding boxes are sketched around the connected components of specified length and width to identify student silhouettes in individual frames (Jalal and Kim 2014; Jalal et al. 2017). Then, a threshold value is set in the range 0 to 1 (Enyedi et al. 2005). Figure 4 presents the silhouette capturing and representation results of hand shaking and kicking behavior interactions.

3.2 Hybrid features algorithms as key body points features

This section provides hybrid feature algorithms which is basically combination of key body points and full-body silhouettes features. In case of key body points features, initially, the boundary of each silhouette and its centroid is identified per frame. These centroids are connected to top boundary point which is identified as head of individual silhouette. Similarly, bottom, left and right extremes are marked. Then, the centroid is connected with boundary pixels at equal intervals to enclose full body silhouette area. Figure 5 presents the overall scenario of key body points retrieval.

Then, the distance between key body points of individual silhouette is estimated as $d(f_1, f_2)$ where first value of the pair belongs to first silhouette f_1 and second value in the pair belongs to other silhouette f_2 is calculated as;

$$d(f_1, f_2) = \sqrt{(f_1x - f_2x)^2 + (f_1y - f_2y)^2} \quad (1)$$

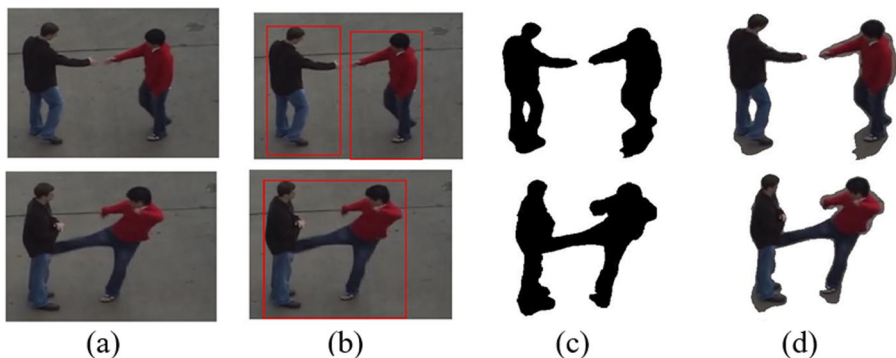


Fig. 4 Silhouette capturing and representation, **a** original frame, **b** silhouette detection, **c** thresholding mechanism, and **d** human silhouettes segmentation of hand shaking and kicking behavior interactions

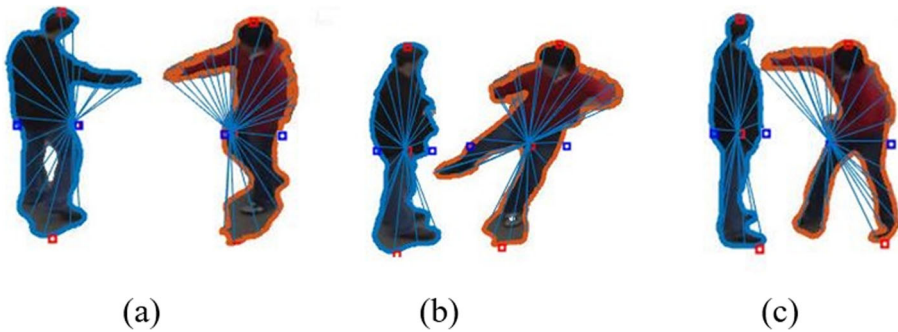


Fig. 5 Examples of key body points retrieval of different human behavior interactions as **a** hand shaking, **b** kicking and **c** punching

where $d(f_1, f_2)$ is the Euclidean distance (Kwang-Kyo 2011; Javed et al. 2010; Sony et al. 2011) between f_1 and f_2 with respect to x and y coordinates. Figure 6 graphically presents the distance between pairs of key body points for hand shaking, kicking and punching behavior interactions.

Based on these distances, spatial relationships are defined as distant and adjacent features.

3.2.1 Distant features

Given the distances between pairs of key body points, a threshold is defined. All the pairs with distance greater than the threshold are taken as distant features and are expressed as;

$$dist(f_1, f_2) \leftrightarrow d(f_1, f_2) \geq threshold \quad (2)$$

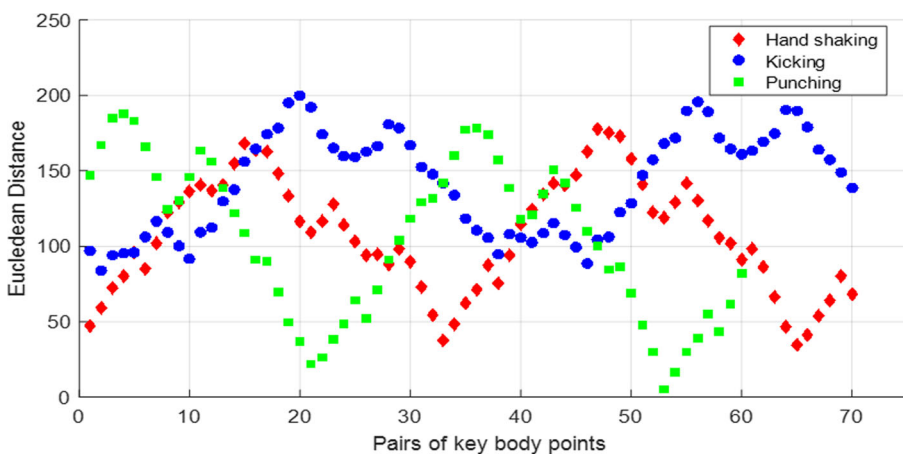


Fig. 6 1D plot of the Euclidean distance between pairs of key body points for hand shaking, kicking and punching behavior interactions

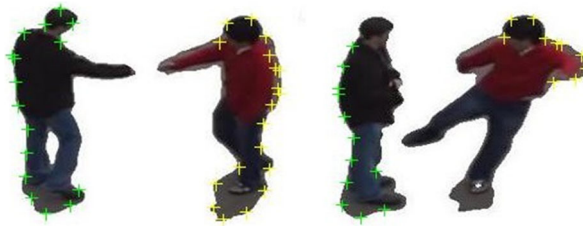


Fig. 7 Examples of Distant features marked for hand shaking and kicking behavior interactions

where $d(f_1, f_2)$ is euclidean distance between f_1 and f_2 . Figure 7 displays distant features marked (i.e., green and yellow + sign) for hand shaking and kicking behavior interactions.

3.2.2 Adjacent features

While, these features $adj(f_1, f_2)$ are said to be adjacent if and only if the distance between key body point pairs is less than the threshold and is measured as;

$$adj(f_1, f_2) \leftrightarrow d(f_1, f_2) \leq threshold \quad (3)$$

where (f_1, f_2) is key body point pair. Figure 8 depicts adjacent features marked (i.e., green and yellow + sign) for hand shaking and kicking behavior interactions.

3.3 Hybrid feature algorithms as Full-body Silhouettes features

Due to similar or closer interactions, it is not sufficient to just restrict with key points features. Therefore, change in intensity of full body silhouettes is measured across all the frames to specify starting and ending frames of each pixel location (Jalal et al. 2012). Based on starting and ending frames, temporal relationships are defined by introducing several robust feature techniques as identical, enclosed and double features. Also histograms of oriented gradient and optical flow are extracted from full body silhouettes to improve performance of the proposed hybrid feature algorithms.

3.3.1 Identical features

In identical features, sequential data of silhouettes pixel values having similar starting and ending frames are extracted and expressed as;

$$iden(f_1, f_2) \leftrightarrow st(f)_1 = st(f_2) \ \& \ end(f_1) = end(f_2) \quad (4)$$

Fig. 8 Examples of adjacent features marked for hand shaking and kicking behavior interactions



where st and end represent starting and ending frames of pixel values f_1 and f_2 , respectively.

3.3.2 Enclosed features

In enclosed features, pixel values that occur during the time frames of other pixel values measured across full video sequences are extracted and expressed as;

$$encl(f_1, f_2) \leftrightarrow st(f_1) > st(f_2) \ \& \ end(f_1) < end(f_2) \quad (5)$$

3.3.3 Double features

In double features, continuous data of silhouettes pixel values that overlap the time frames of other pixel values are extracted and expressed as;

$$dbl(f_1, f_2) \leftrightarrow st(f_2) < end(f_1) \ \& \ st(f_2) > st(f_1) \quad (6)$$

These features are visually drawn (as shown in Fig. 9) and marks the location of temporal features over single frame for hand shaking and kicking behavior interactions, respectively.

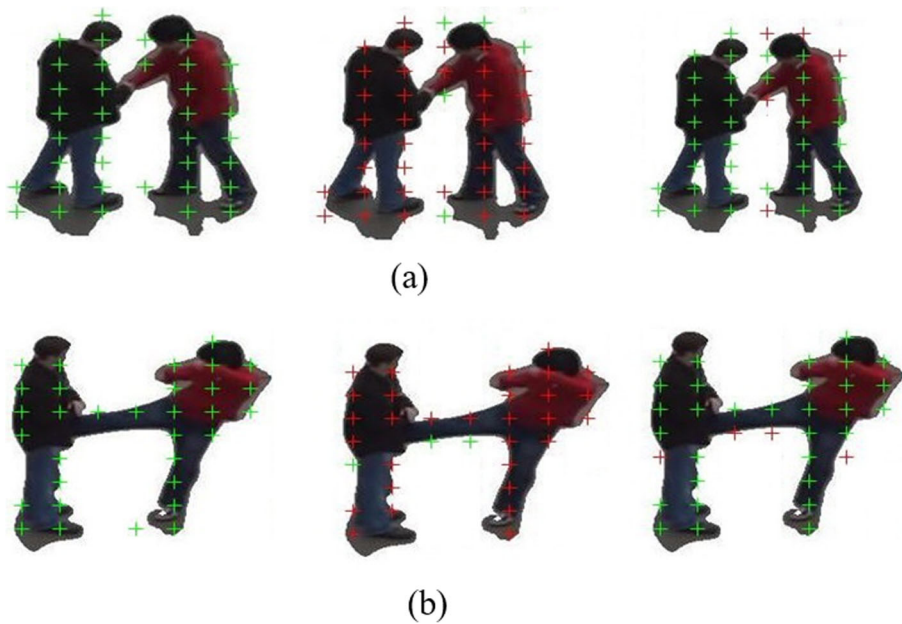


Fig. 9 Temporal relationships, **a** identical (left), enclosed (middle) and double (right) features for hand shaking behavior interaction, **b** identical (left), enclosed (middle) and double (right) features for kicking behavior interaction

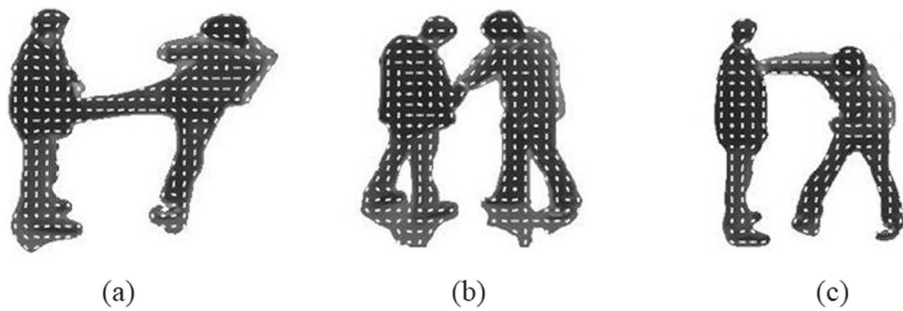


Fig. 10 Examples of HOG features over **a** kicking, **b** handshake and **c** punching behavior interactions

3.3.4 HOG features

In HOG features, occurrences of gradient orientation in segmented silhouettes is counted using overlapping local contrast normalization on a dense grid of uniformly spaced cells (as shown in Fig. 10)

3.3.5 HOF features

In HOF features, optical flow objects are created and optical flow is estimated using X-Y coordinate values, orientation and magnitude. Figure 11 shows the results of HOF extraction over kicking, handshake and punching interactions of UT-Interaction dataset.

3.4 Co-occurrence matrix computation

To overcome long execution time, robust spatio-temporal features are represented by a co-occurrence matrix (Li and Lu 2009; Jalal et al. 2014; Walker et al. 2002) that distributes co-occurring features at a given offset as;

$$C_{m,n} = \frac{1}{N} \sum_{f=1}^F (\sigma(S_{1m}, S_{2n})) \quad (7)$$

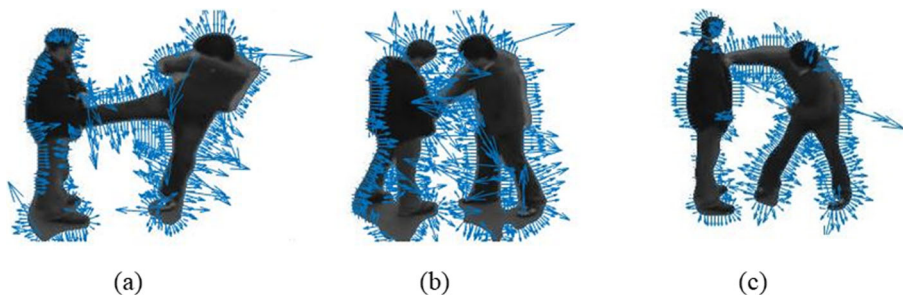


Fig. 11 Examples of HOF features over **a** kicking, **b** handshake and **c** punching behavior interactions

where S_{1m} is the m^{th} features of first silhouette and S_{2n} is the n^{th} features of second silhouette of similar image (Fan and Wang 2004; Barnard et al. 2008). S_{1m} and S_{2m} co-occur when difference between their time-frames is less than a threshold (Maric and Kolarov 2002; Wang et al. 2008; Tang et al. 2008). While, N is the normalization term and F represents the total number of frames.

3.5 HIR via artificial neural network

Finally, these mixed co-occurrence matrices are needed to be properly modelled, trained and tested (Lynch and Willett 2002; Wang et al. 2016; Zhang et al. 2012) using reasonable recognizer engine (Siswanto et al. 2015; Xu and Lei 2008; Huang 2010). Therefore, extracted spatio-temporal features are fed to artificial neural network (Turcanik 2010; Maa and Schanblatt 1992) which has a transfer function T_j to compute net input as;

$$T_j = \sum_i (w_{i,j} \times x_i + b_j) \quad (8)$$

where $w_{i,j}$ represents the weights at links of the neural network that connect its adjacent layers, x_i represents input features and b_j is the added bias. In addition,

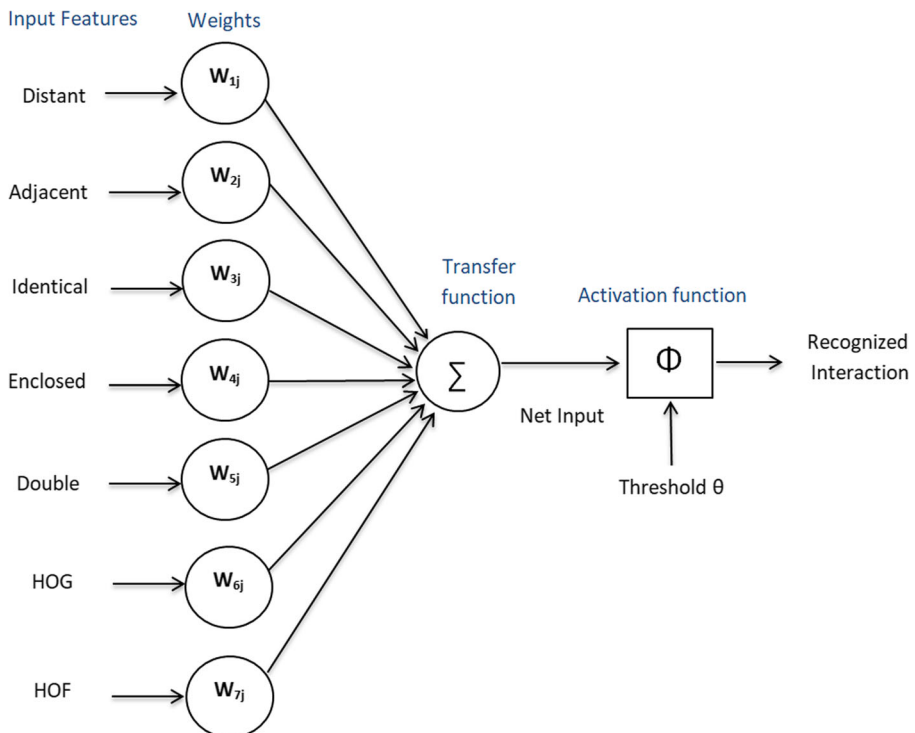


Fig. 12 Structure and probabilistic parameters used in artificial neural network over different behavior interactions

softmax activation function simulates the complex behavior of human cognition by non-linearly mapping the co-occurring spatio-temporal features during interaction class (Lavalle and Rodriguez 2007). The outcome of the softmax function $\sigma(T)_j$ gives probability distribution over k possible behavior interactions as;

$$\sigma(T)_j = \frac{e_j^T}{\sum_{i=1}^k (e_i^T)}, j = 1, \dots \quad (9)$$

where T_j represents the summation of input features multiplied by corresponding weights (Tsang et al. 2003; Sima 2017) as net input from transfer function. Figure 12 shows the overall concept of artificial neural network applied over different behavior interactions.

4 Experimental Results

In this paper, we have proposed human interaction recognition technique for student behavior mining in the E-learning environment. Particularly online martial arts' training and classroom behaviors are focused in our work. The idea of video based learning for students while adding interactivity and providing feedbacks using interaction recognition techniques and classroom behaviors dataset is our major contribution as innovative technologies in education. To evaluate the performance of proposed method over UT-Interaction dataset (Ryoo and Aggarwal 2009), we performed three different experiments. Each experiment validates SBR, while varying in division of training and validation sets. However, to test the effectiveness of proposed idea over classroom behaviors, we visited different classrooms randomly at educational Centers (i.e., Kindergarten, Schools and universities). We met trained ICT instructors and conducted video based interactive learning experiments with students. Students were shown different videos of daily manners, good habits and student behaviors in classrooms. Then those students were asked to perform the same interactions under E-learning environment. Their behaviors were recorded and given to proposed SBR system for recognition. In addition, this section presents dataset explanation, recognition accuracy and comparison of proposed method with other state-of-the-art recognition methods.

mean recognition accuracy.

4.1 UT-Interaction dataset explanation

UT-Interaction dataset (Ryoo and Aggarwal 2009) contains twenty video sequences of six different behavior interactions between two persons. These behavior interactions include handshaking, hugging, kicking, punching, pushing and pointing. Handshaking and hugging are considered positive behaviors. Kicking, punching and pushing are negative while pointing is a neutral behavior. Several actors are appeared with 15 different clothing colors having the resolution of 720x480 at 30fps. The dataset is equally divided into two sets as set 1 and set 2. Set 1 are taken in a parking lot with slightly different zoom rate and their backgrounds are mostly static.



Fig. 13 Examples of six different behavior interactions of the UT-Interaction dataset

While, set 2 are taken on a lawn in a windy day with slightly moving background and more camera jitters. The dataset is focused on teaching online action karate to the kindergarten students. Figure 13 shows the six different behavior interactions of the UT-Interaction dataset.

4.2 Experiment 1: One-third testing validation test

For the experimentation, one third of the samples are used as testing and the rest is used as training i.e out of 10 training instances per interaction, 6 are used for training while 4 have undergone testing. The experiment is performed on set 1 (parking as background) and set 2 (lawn as background) separately. Tables 1 and 2 show the confusion matrix of six different behavior interactions having **91.6%** and **83.3%**

Table 1 Confusion matrix for one-third testing validation test Of UT-Interaction set 1

	Hand shaking	Hugging	Kicking	Punching	Pushing	Pointing
Hand shaking	0.75	0.00	0.00	0.25	0.00	0.00
Hugging	0.00	1.0	0.00	0.00	0.00	0.00
Kicking	0.00	0.00	1.00	0.00	0.00	0.00
Punching	0.00	0.00	0.00	1.00	0.00	0.00
Pushing	0.25	0.00	0.00	0.00	0.75	0.00
Pointing	0.00	0.00	0.00	0.00	0.00	1.0

Table 2 Confusion matrix for one-third testing validation test of UT-Interaction set 2

	Hand shaking	Hugging	Kicking	Punching	Pushing	Pointing
Hand shaking	1.00	0.00	0.00	0.00	0.00	0.00
Hugging	0.00	1.00	0.00	0.00	0.00	0.00
Kicking	0.00	0.00	1.00	0.00	0.00	0.00
Punching	0.25	0.00	0.00	0.5	0.25	0.00
Pushing	0.25	0.25	0.00	0.00	0.5	0.00
Pointing	0.00	0.00	0.00	0.00	0.00	1.00

4.3 Experiment 2: Two-Third Testing Validation Test

In the second experiment, two third of the samples undergo testing and rest are used for training i.e. out of 10 training instances per interaction, 6 are used for testing while 4 have undergone training. The experiment is performed separately on set 1 and set 2. Tables 3 and 4 present the confusion matrix of six different behavior interactions having **83.3%** and **75%** mean recognition accuracy.

4.4 Experiment 3: 10-fold Cross Validation Test

In our last experiment, we performed 10-fold cross validation test i.e. 10 iterations are performed while each instance has undergone testing. Mean recognition rate is calculated by taking the average of performance results over 10 iterations. Tables 5 and 6 show the confusion matrix of six different behavior interactions having **88.3%** and **80.0%** mean recognition accuracy. Figures 14 and 15 show the visual representation of experimental results over set 1 and set 2 of UT-Interaction dataset respectively.

Furthermore, to verify the correctness of detected moves, we have compared the predicted interactions with the ground truth labels. If the comparison results are true, detected moves are correctly identified and vice versa. Then the ratio of

Table 3 Confusion matrix for two-third testing validation test of UT-Interaction set 1

	Hand shaking	Hugging	Kicking	Punching	Pushing	Pointing
Hand shaking	1.00	0.00	0.00	0.00	0.10	0.00
Hugging	0.00	0.83	0.00	0.17	0.00	0.00
Kicking	0.00	0.00	0.67	0.33	0.00	0.00
Punching	0.165	0.00	0.00	0.67	0.165	0.00
Pushing	0.17	0.00	0.00	0.00	0.83	0.00
Pointing	0.00	0.00	0.00	0.00	0.00	1.00

Table 4 Confusion matrix for two-third testing validation test of UT-Interaction set 2

	Hand shaking	Hugging	Kicking	Punching	Pushing	Pointing
Hand shaking	0.83	0.00	0.00	0.00	0.17	0.0
Hug	0.00	0.67	0.00	0.33	0.00	0.00
Kick	0.00	0.00	1.00	0.00	0.00	0.00
Punch	0.00	0.00	0.33	0.5	0.17	0.00
Push	0.50	0.00	0.00	0.00	0.5	0.00
Point	0.00	0.00	0.00	0.00	0.00	1.00

Table 5 Confusion matrix for cross validation test of UT-Interaction Set 1

	Hand shaking	Hugging	Kicking	Punching	Pushing	Pointing
Hand shaking	1.0	0.0	0.0	0.0	0.0	0.0
Hugging	0.0	0.7	0.0	0.1	0.2	0.0
Kicking	0.0	0.0	1.0	0.0	0.0	0.0
Punching	0.1	0.0	0.0	0.8	0.1	0.0
Pushing	0.1	0.1	0.0	0.0	0.8	0.0
Pointing	0.0	0.0	0.0	0.0	0.0	1.0

Table 6 Confusion matrix for cross validation test of UT-Interaction Set 2

	Hand shaking	Hugging	Kicking	Punching	Pushing	Pointing
Hand shaking	0.7	0.2	0.0	0.1	0.0	0.0
Hugging	0.0	0.7	0.0	0.3	0.0	0.0
Kicking	0.0	0.0	0.8	0.2	0.0	0.0
Punching	0.1	0.0	0.0	0.7	0.2	0.0
Pushing	0.0	0.1	0.0	0.0	0.9	0.0
Pointing	0.0	0.0	0.0	0.0	0.0	1.0

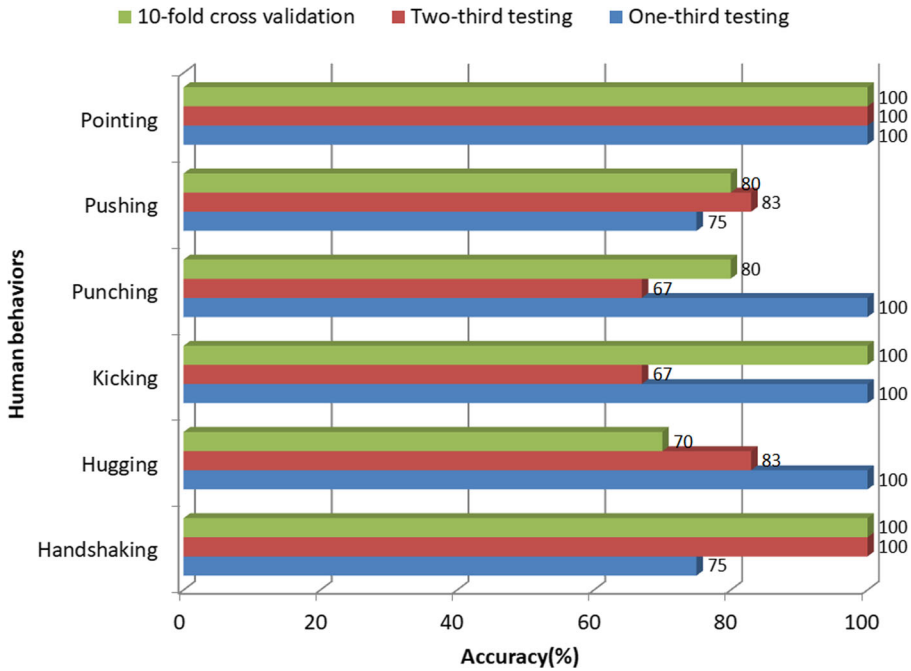


Fig. 14 Visual comparison of experimental results over set 1 of UT-Interaction dataset

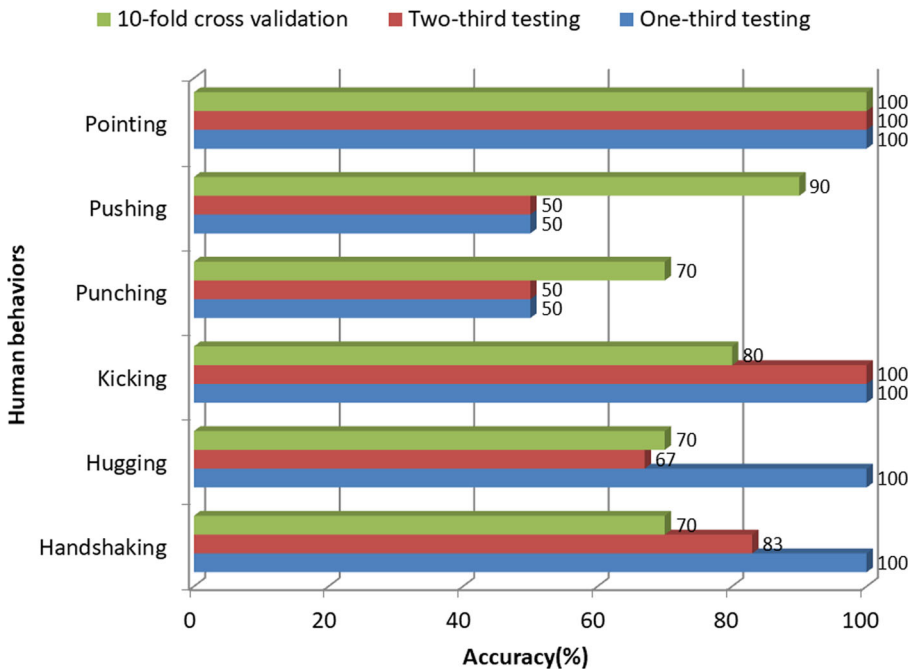


Fig. 15 Visual comparison of experimental results over set 2 of UT-Interaction dataset

Table 7 Comparison of recognition performance over different experiments as one-third, two-third and cross validation testing

Experiments	Set 1 (%)	Set 2 (%)	Mean (%)
One-third testing	91.6	83.3	87.45
Two-third testing	83.3	75	79.15
Cross Validation	88.3	80	84.15

correctly identified moves with total tested moves is calculated to give accuracy. Table 7 presents the mean accuracy of three experiments performed on UT-Interaction dataset and Table 8 compares the recognition accuracy of the proposed hybrid spatio-temporal features method with that of the state-of-the-art methods (Houda and Yannick 2014; Ryoo and Aggarwal 2009).

From the experiments, it is observed that 100 percent recognition results are obtained for the point interaction over each experiment but some confusion are still observed between similar interactions like punching, pushing and handshake. Comparison on performance results over set 1 and set 2 demonstrates a decrease in performance over set 2 due to composite background situations. And finally a comparison between the three experiments show that maximum performance is achieved over one-third testing while lowest performance is achieved over two-third testing due to insufficient training instances.

4.5 Classroom behaviors dataset explanation

Classroom behaviors dataset contains ten video sequences of twelve different behavior interactions between student-student or student-object. These behavior interactions include read, write, teach, use computers, stand up, sit down, raise hands, handshake, punch, kick, play and swing. Several students are appeared in uniform clothing. Scenes are captured in indoor classroom and outdoor playground environments. The dataset is focused to teach students good and bad classroom manners. Figure 16 shows the twelve different behavior interactions of the classroom behaviors dataset.

4.6 Classroom behaviors recognition results

To evaluate the performance of proposed SBR system over classroom behaviors dataset, we performed 10-fold cross validation testing. Table 9 shows the recognition

Table 8 Comparison of recognition accuracy on UT-Interaction dataset with other state of the art methods

Methods	Recognition accuracy (Set 1)%	Recognition accuracy (Set 2) %
BoW (Houda and Yannick 2014)	58.2	48.3
HIR based on co-occurrence of visual words (Houda and Yannick 2014)	40.63	66.67
STR Match kernel (Ryoo and Aggarwal 2009)	70.8	70.8
Proposed Approach	91.6	83.3



Fig. 16 Examples of twelve different behavior interactions of the classroom behaviors dataset where first figure shows visual shown to the students while second figure shows interactions performed by the students

accuracies of twelve different student behaviors having mean recognition rate of **73.3%**.

4.7 Survey results

After experiments, we collected the views of both instructors and students about video based interactive learning using questionnaire based survey. Two kind of questionnaires were designed containing 5 questions with four options each as strongly agree, agree, neither agree nor disagree and disagree. One questionnaire was for the instructors to evaluate their ease in teaching and effectiveness of adopting video based interactive learning in classrooms. Second was for the students to analyze their interests, level of learning and compatibility with technology. A total of 20 instructors

Table 9 Recognition accuracies on twelve different interactions of the classroom behaviors dataset

Classroom behaviors	Accuracy %
Read	60
Write	70
Teach	60
Use computers	70
Stand up	80
Sit down	80
Raise hands	70
Handshake	90
Punch	90
Kick	90
Play	60
Swing	60
Mean	73.3

Survey questions for instructors about Video based Interactive Learning (VIL)	Strongly Agree	Agree	Neither agree nor disagree	Disagree
1. VIL is more effective than traditional teaching practices	16	4		
2. It is difficult to tackle students under VIL environment	2	8	10	
3. VIL bring ease for teachers in delivering the lecture	4	14	2	
4. It is difficult to incorporate VIL in our classrooms	12	5	3	
5. Despite of the challenges, VIL is worth introducing in our classrooms	5	12	3	

Fig. 17 Survey questions from instructors and their quantitative results

and 40 students participated in our survey. Figures 17 and 18 depict survey questions for instructors and students about video based interactive learning and their quantitative results.

Survey questions for instructors about Video based Interactive Learning (VIL)	Strongly Agree	Agree	Neither agree nor disagree	Disagree
1. VIL is more interesting than traditional teaching practices	20	15	5	
2. It is easy to learn and practice	18	2	20	
3. More of such lessons should be added in classrooms	40			
4. Do you use computers at home?		20		20
5. Would you practice VIL at home?		22	8	

Fig. 18 Survey questions from students and their quantitative results

Based on the survey results, it is concluded that 85% of the instructors appreciated video based online interactive learning and claimed that such practices should be included in the classrooms. While 15% of the instructors were neutral about introducing e-learning techniques in classrooms due to expensive environmental settings. However students were all very happy and excited to experience video based learning. Majority of the students participated in the experiments and claimed that they need more of such lessons. But 50% of the students stated that they do not use computers at home and hence face difficulty interacting with technology.

5 Conclusion

In this paper, an approach for behavior mining is proposed using hybrid spatio-temporal feature algorithms. The idea of extracting spatio-temporal features from full body silhouettes is extended to key body points. Performance of the proposed approach is measured on UT-Interaction and classroom behaviors dataset with the focus on Martial Arts training to kindergarten students for character development, etiquettes, and self-defense. Experimental results validate the success of our methodology, while a comparison with other state-of-the-art recognition methods suggests that our model distinguishes critical behavior interactions with complex environments better than the previous systems.

In the future, we are planning to introduce virtual invigilators in E-quizzing testing. Student interactions like cheating and exchanging materials will be kept in focus while improving the performance of our proposed SBR system. Another direction where we are looking forward is E-gaming. Players' physical interactions will be recognized to produce responsive feedbacks in online games. Also, we aim at definition of skeleton model from occluded body parts and extraction of angular and geometric features to add flexibility into proposed model.

Compliance with Ethical Standards

Conflict of interests The authors declare that they have no conflict of interest.

References

- Buyts, K., Cagniard, C., Baksheev, A., Laet, T.-D., Schutter, J.D., Pantofaru, C. (2014). An adaptable system for RGB-D based human body detection and pose estimation. *Journal of visual communication and image representation*, 25, 39–52.
- Oberg, J., Eguro, K., Bittner, R., Forin, A. (2012). Random decision tree body part recognition using FPGAS. In: Proceedings of international conference on field programmable logic and applications, pp. 330–337.
- Jalal, A., & Zeb, M.A. (2008). Security enhancement for e-learning portal. *International Journal of Computer Science and Network Security*, 8(3), 41–45.
- Kanungo, T., Mount, D.M., Netanyahu, N.S., Piatko, C.D., Silverman, R., Wu, A.Y. (2002). An efficient k-means clustering algorithm: analysis and implementation. *IEEE Transaction on Pattern Analysis and Machine Intelligence*, 24(7), 881–892.
- Yang, X., & Tian, Y. (2014). Super normal vector for activity recognition using depth sequences. In: Proceedings of CVPR conference, Columbus, pp. 804–811.

- Sehar, R., Mahmood, M., Yousaf, S., Khatoon, H., Khan, S., Moqurab, S.A. (2018). An Investigation on Students Speculation towards Online Evaluation. In: Proceedings of 11th International Conference on Assessments and Evaluation on global south.
- Yang, X., & Tian, Y. (2012). Eigenjoints-based action recognition using naive-bayes-nearest-neighbor. In: Proceedings of CVPR conference, Providence, RI, pp 14–19.
- Jalal, A., Kim, Y., Kim, D. (2014). Ridge body parts features for human pose estimation and recognition from RGB-D video data. In: Proceedings of the IEEE international conference on computing, communication and networking technologies.
- Muller, M., & Roder, T. (2006). Motion templates for automatic classification and retrieval of motion capture data. In: Proceedings of ACM symposium on computer animation, Austria, pp. 137–146.
- Mahmood, M., Jalal, A., Evans, H.A.I.n.p.ress. (2018). Facial expression recognition in image sequences using 1D transform and gabor wavelet transform. In: Proceedings of international conference on applied and engineering mathematics.
- Fatahi, S., Shabanali-Fami, F., Moradi, H. (2018). An empirical study of using sequential behavior pattern mining approach to predict learning styles. *Journal of Education and Information Technologies*, 23(4), 1427–1445.
- Aissaoui, O., Madani, Y., Oughdir, L., Alloui, Y. (2018). A fuzzy classification approach for learning style prediction based on web mining technique in e-learning environments. *Journal of Education and Information Technologies*, pp. 1–17.
- Zhao, X., Li, X., Pang, C., Wang, S. (2013). Human action recognition based on semi-supervised discriminant analysis with global constraints. *Neurocomputing*, 105, 45–50.
- Jalal, A., Sharif, N., Kim, J.T., Kim, T.S. (2013). Human Activity Recognition via Recognized Body Parts of Human Depth Silhouettes for Residents Monitoring Services at Smart Home. *Indoor and Built Environment*, 22, 271–279.
- Houda, K., & Yannick, F. (2014). Human interaction recognition based on the co-occurrence of visual words. In: Proceedings of CVPR conference, pp. 455–460.
- Ryoo, M.S., & Aggarwal, J.K. (2009). Spatio-temporal relationship match: video structure comparison for recognition of complex human activities. In: Proceedings of ICCV, pp.1593–1600.
- Berlin, S.J., & John, M. (2016). Human interaction recognition through Deep Learning Network. In: Proceedings of IEEE International Carnahan conference on security technology.
- Chattopadhyay, C., & Das, S. (2016). Supervised framework for automatic recognition and retrieval of interaction: a framework for classification and retrieving videos with similar human interactions. *IET Computer Vision*, 10, 220–227.
- Zhan, S., & Chang, I. (2014). Pictorial structures model based human interaction recognition. In: Proceedings of ICMLC, pp. 862–866.
- Hwang, G.-J., Yang, T.-C., Tsai, C.-C., Yang, J.H. (2009). A context-aware ubiquitous learning environment for conducting complex science experiments. In: Computers and Education, Volume 53 (2).
- Lu, T., Zhang, S., Hao, Q., Yang, J.H. (2012). Activity Recognition in Ubiquitous Learning Environment. In: Journal of advances in information technology, Volume 3 (1).
- Chen, K., Yue, G., Yu, F., Shen, Y., Zhu, A. (2007). Research on speech emotion recognition system in E-learning. In *Lecture notes in computer science*, Vol. 4489. Berlin: Springer.
- Kowalewski, W., Koodziejczak, B., Roszak, M., Ren-Kurc, A. (2013). Gesture recognition technology in education. In: Distance learning, simulation and communication, pp. 113–120.
- Zaletelj, J., & Košir, A. (2017). Predicting students' attention in the classroom from Kinect facial and body features. In: EURASIP journal on image and video processing.
- Sabanc, O., & Bulut, S. (2018). The Recognition and Behavior Management of Students With Talented and Gifted in an Inclusive Education Environment. In: Journal of Education and Training Studies, Volume 6 (6).
- Gaschler, A., Jentzsch, S., Giuliani, M., Huth, K., Ruiter, J., Knoll, A. (2012). Social behavior recognition using body posture and head pose for human-robot interaction. In: Proceedings of IEEE/RSJ international conference on intelligent robots and systems, pp. 2128–2133.
- Fujii, T., Lee, J., Okamoto, S. (2014). Gesture Recognition System for Human-Robot Interaction and its application to robotic service task. In: Proceedings of international multicongference of engineers and computer scientists, pp. 63–68.
- Babiker, M., Khalifa, O., Htyke, K., Hassan, A., Zaharadeen, M. (2017). Automated daily human activity recognition for video surveillance using neural network. In: Proceedings of IEEE 4th International Conference on Smart Instrumentation, Measurement and Application, pp. 1–5.

- Gkioxari, G., Girshick, R., Dollár, P., He, K. (2018). Detecting and recognizing human-object interactions. In: Proceedings of computer vision and pattern recognition.
- Shen, L., Yeung, S., Hoffman, J., Mori, G., Fei, L. (2018). Scaling human-object interaction recognition through zero-shot learning. In: Proceedings of IEEE winter conference on applications of computer vision, pp. 1568–1576.
- Cho, N., Park, S., Park, J., Park, U., Lee, S. (2017). Compositional interaction descriptor for human interaction recognition. *Neurocomputing*, pp. 169–181.
- Kong, Y., Liang, W., Dong, Z., Jia, Y. (2014). Recognizing human interactions from videos by a discriminative model. *IET Computer Vision*, 8, 277–286.
- Ma, L., Liu, J., Wang, J. (2009). A improved silhouette tracking approach integrating particle filter with graph cuts. In: Proceedings of ICCV, pp.1593–1600.
- Jalal, A., Kim, J.T., Kim, T.-S. (2012). Human activity recognition using the labeled depth body parts information of depth silhouettes. In: Proceedings of the 6th international symposium on sustainable healthy buildings, pp. 1–8.
- Milanfar, P. (2012). A tour of modern image filtering: New insights and methods, both practical and theoretical. *IEEE signal processing magazine*, 30, 106–128.
- Jalal, A., & Kim, Y. (2014). Dense depth maps-based human pose tracking and recognition in dynamic scenes using ridge data. In: Proceedings of the IEEE international conference on advanced video and signal-based surveillance, pp. 119–124.
- Jalal, A., Kim, Y.-H., Kim, Y.-J., Kamal, S., Kim, D. (2017). Robust human activity recognition from depth video using spatiotemporal multi-fused features. *Pattern recognition*, 61, 295–308.
- Enyedi, B., Konyha, L., Fazekas, K. (2005). Threshold procedures and image segmentation. In: Proceedings of the IEEE international symposium ELMAR, pp. 119–124.
- Kwang-Kyo, H.-S. (2011). Distance-based formation control using euclidean distance dynamics matrix: Three-agent case. In: Proceedings of american control conference, pp. 4810–4815.
- Javed, J., Yasin, H., Ali, S. (2010). Human movement recognition using euclidean distance: A tricky approach. In: Proceedings of 3rd international congress on image and signal processing.
- Sony, A., Ajith, K., Thomas, K., Thomas, T., Deepa, P.L. (2011). Video summarization by clustering using euclidean distance. In: Proceedings of international conference on signal processing, communication, Computing and Networking Technologies.
- Jalal, A., Kim, J.T., Kim, T.-S. (2012). Development of a life logging system via depth imaging-based human activity recognition for smart homes. In: Proceedings of the international symposium on sustainable healthy buildings, pp. 91–95.
- Li, Q., & Lu, W. (2009). A histogram descriptor based on co-occurrence matrix and its application in cloud image indexing and retrieval. In: Proceedings of 5th international conference on intelligent information hiding and multimedia signal processing.
- Jalal, A., Kamal, S., Kim, D. (2014). A depth video sensor-based life-logging human activity recognition system for elderly care in smart indoor environments. *Sensors*, 14, 11735–11759.
- Walker, R.F., Jackway, P.T., Longstaff, I.D. (2002). Recent developments in the use of the co-occurrence matrix for texture recognition. In: Proceedings of 13th international conference on digital signal processing.
- Fan, B., & Wang, Z. (2004). Pose estimation of human body based on silhouette images. In: Proceedings of international conference on information acquisition.
- Barnard, M., Matilainen, M., Heikkilä, J. (2008). Body part segmentation of noisy human silhouette images. In Proceedings of IEEE international conference on multimedia and expo.
- Maric, S.V., & Kolarov, A. (2002). Threshold based admission policies for multi-rate services In: the DECT system. In: Proceedings of 6th international symposium on personal, indoor and mobile radio communications.
- Wang, W., Qin, Z., Rong, S., Xingfu, S.R. (2008). A kind of method for selection of optimum threshold for segmentation of digital color plane image. In: Proceedings of 9th international conference on computer-aided industrial design and conceptual design.
- Tang, X., Pang, Y., Zhang, H., Zhu, W. (2008). Fast image segmentation method based on threshold. In: Proceedings of Chinese control and decision conference.
- Lynch, R., & Willett, P. (2002). Classification with a combined information test. In: Proceedings of IEEE international conference on acoustics, speech, and signal processing.
- Wang, J., Wang, S., Cui, Q., Wang, Q. (2016). Local-based active classification of test report to assist crowdsourced testing. In: Proceedings of IEEE international conference on automated software engineering, pp. 190–201.

- Zhang, J., Chen, C., Xiang, Y., Zhou, W. (2012). Semi-supervised and compound classification of network traffic. In: Proceedings of international conference on distributed computing systems workshops, pp. 617–62.
- Siswanto, A., Nugroho, A., Galinium, M. (2015). Implementation of face recognition algorithm for biometrics based time attendance system. In: Proceedings of International Conference on ICT for Smart Society.
- Xu, G., & Lei, Y. (2008). A new image recognition algorithm based on skeleton. In: Proceedings of IEEE world congress on computational intelligence.
- Huang, H. (2010). A simplified image recognition algorithm based on simple scenarios. In: Proceedings of international conference on computational intelligence and software engineering.
- Turcanik, M. (2010). Network routing by artificial neural network. Military communications and information systems conference.
- Maa, C.Y., & Schanblatt, M.A. (1992). A two-phase optimization neural network. *IEEE Transactions on Neural Networks*, vol. 3.
- Lavalle, M., & Rodriguez, G. (2007). Feature selection with interactions for continuous attributed and discrete class. In: Proceedings of electronics, robotics and automotive mechanics conference.
- Tsang, E.C.C., Huang, D.M., Yeung, D.S., Lee, J.W.T., Wang, X.Z. (2003). A weighted fuzzy reasoning and its corresponding neural network. In: Proceedings of IEEE international conference on systems, man and cybernetics.
- Sima, J. (2017). Neural networks between integer and rational weights. In: Proceedings of international joint conference on neural networks.
- Ryoo, M.S., & Aggarwal, J.K. (2009). Spatio-temporal relationship match: video structure comparison for recognition of complex human activities. In: Proceedings of IEEE international conference on computer vision.

Publisher's note Springer Nature remains neutral with regard to jurisdictional claims in published maps and institutional affiliations.